

Factor each completely. Show your work for partial credit.

|                                |                              |                             |                     |
|--------------------------------|------------------------------|-----------------------------|---------------------|
| 1. $25x^2 - 49y^2$             |                              | 2. $6x^3 - 18x^2 + 5x - 15$ |                     |
| Factored Form                  | $(5x + 7y)(5x - 7y)$         | Factored Form               | $(6x^2 + 5)(x - 3)$ |
| 3. $24x^5y^6z^4 + 54x^8y^4z^9$ |                              | 4. $16x^2 + 72x + 81$       |                     |
| Factored Form                  | $6x^5y^4z^4(4y^2 + 9x^3z^5)$ | Factored Form               | $(4x + 9)^2$        |
| 5. $x^2 - 5x - 14$             |                              | 6. $9x^2 + 21x + 10$        |                     |
| Factored Form                  | $(x + 2)(x - 7)$             | Factored Form               | $(3x + 2)(3x + 5)$  |

|                             |                      |                             |                        |
|-----------------------------|----------------------|-----------------------------|------------------------|
| 7. $2x^2 + x - 36$          |                      | 8. $x^2 + 7x - 30$          |                        |
| <b>Factored Form</b>        | $(2x + 9)(x - 4)$    | <b>Factored Form</b>        | $(x + 10)(x - 3)$      |
| 9. $x^6 - 25$               |                      | 10. $x^2 + 20x + 99$        |                        |
| <b>Factored Form</b>        | $(x^3 + 5)(x^3 - 5)$ | <b>Factored Form</b>        | $(x + 9)(x + 11)$      |
| 11. $5x^2 - 4x + 20x^3 - 1$ |                      | 12. $24x^2y - 16xy + 56y^3$ |                        |
| <b>Factored Form</b>        | $(5x^2 - 1)(4x + 1)$ | <b>Factored Form</b>        | $8y(3x^2 - 2x + 7y^2)$ |

|                                                                 |                                                                                 |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| 13. $x^2 - 10x + 21$                                            | 14. $8x^3y^2 + 56x^7y^5 + 32x^6y^4$                                             |
| <div>Factored Form</div> <div><math>(x - 7)(x - 3)</math></div> | <div>Factored Form</div> <div><math>8x^3y^2(1 + 7x^4y^3 + 4x^3y^2)</math></div> |
| 15. $x^2 + 16x + 64$                                            | 16. $8x^2 + 22x - 21$                                                           |
| <div>Factored Form</div> <div><math>(x + 8)^2</math></div>      | <div>Factored Form</div> <div><math>(4x - 3)(2x + 7)</math></div>               |

17. An expression is shown.

$$8x^3 - 12x^2 - 2x + 3$$

Select all of the equivalent expressions.

- ☐  $2x(4x^2 - 6x - 1)$
- ☐  $2x^2(4x - 6x - 1)$
- ☐  $4x^2(2x - 3) - 1(2x + 3)$
- ☒  $4x^2(2x - 3) - 1(2x - 3)$
- ☒  $(2x - 3)(2x - 1)(2x + 1)$